

Project 101: BigOutsource

Employee Information

Management System (EIMS)

Welcome to Project 101! This document is a beginner-friendly, plain-English overview of the BigOutsource Employee Information Management System (EIMS). Whether you are a new team member, an HR specialist, an IT administrator, a project manager, or an executive stakeholder, this guide will help you quickly understand what the platform does, how it works under the hood, and how to navigate its key workflows.

1. Executive Summary & Core Mission

What is EIMS?

The BigOutsource Employee Information Management System (EIMS) is a centralized workforce operational platform built specifically for BigOutsource. Before EIMS, employee records, IT provisioning steps, workstation assignments, and evaluation schedules were managed across disconnected spreadsheets and email threads. EIMS consolidates all workforce operations into a secure, single source of truth.

Core Business Problem Solved

1. **Spreadsheet Fragmentation:** Eliminates reliance on outdated, duplicated Excel sheets across HR and IT departments.
2. **Provisioning Bottlenecks:** Streamlines handoffs between HR onboarding and IT asset provisioning (assigning emails, LMS accounts, workstations, and software licenses).
3. **Data Security & Secret Protection:** Replaces plain-text password sharing with encrypted database storage, strict role-based access control, and audit logging.
4. **Missed Evaluation Milestones:** Automates notifications for employee evaluation dates (1st, 3rd, 5th, 6th month, and annual anniversary checks).

5. **Equipment & License Tracking:** Tracks hardware inventory (laptops, desktops, monitors) and software licenses linked directly to individual employees.

Target End-Users

- **HR Managers & Recruiters:** Manage onboarding, employment status changes, evaluation dates, personal records, and birthday alerts.
- **IT Administrators:** Issue corporate emails, LMS accounts, workstations, REMOTE ID connection codes, antivirus (ESET) and activity monitoring (ActivityWatch) statuses, and disk encryption keys.
- **Operations Leads & Department Managers:** View real-time DEPARTMENT/CAMPAIGN rosters, site occupancy, and team structures.
- **Executive Leadership & Directors:** Access high-level workforce analytics, security compliance audits, and custom data exports.
- **System Administrators:** Control user accounts, grant granular permissions, edit custom roles, and monitor system audit trails.

Main Feature Modules

- **Directory & EMPLOYEE INFORMATION:** Central hub for employee records, contact info, government IDs (SSS, TIN, PhilHealth, Pag-IBIG), and employment status.
 - **IT Credentials & Security Provisioning:** Dedicated workspace for IT teams to assign corporate credentials, software keys, and track security tool compliance.
 - **Hardware Asset Inventory:** Complete tracking of company-owned hardware assets, device serial numbers, assignment histories, and maintenance states.
 - **DEPARTMENT/CAMPAIGN & Site Organization:** Categorization of workforce units across multiple physical sites (HQ - San Pablo, Candelaria, WFH, Hybrid) and operational departments/campaigns.
 - **Staging & Bulk Imports:** Spreadsheet importer that validates raw CSV/Excel files, highlights potential duplicate records, and stages data before importing.
 - **Workforce Analytics & Export Engine:** Customizable reports and one-click Excel downloads for master rosters, IT asset logs, site occupancy, and terminations.
 - **Automated Schedulers & Notifications:** Daily background checks that send birthday alerts, evaluation reminders, and real-time in-app notification badges.
 - **System Security & Audit Logging:** Multi-Factor Authentication (MFA), single-session enforcement, and immutable audit logs capturing every record modification.
-

2. High-Level Architecture (Plain English)

Even if you don't have a background in software engineering, understanding how EIMS works is simple. Think of the system as a modern 3-tier building:



Key Components Explained

1. The Frontend (Client Application):

- What it is: The interactive web application you see in your Google Chrome, Edge, or Safari browser.
 - How it works: Built using React 19 and Vite, it loads instantly, provides interactive search filters, renders smooth animations, and automatically updates data without requiring full page refreshes.
2. The Backend API (Server Application):
- What it is: The background server running on Node.js and Express.
 - How it works: Acts as the security guard and logic engine. When you click "Save Employee" in the browser, the frontend sends a secure request to the backend. The backend checks if your account has permission to perform that action, validates the data, updates the database, and records an entry in the system audit log.
3. The Database (Storage Vault):
- What it is: A high-performance PostgreSQL relational database managed via Prisma ORM.
 - How it works: Organizes information into neat, interconnected tables (user_profiles, employees, accounts, devices, audit_logs).
4. Realtime Engine & Background Workers:
- Socket.io (WebSockets): Maintains an instant live connection to display who is currently online in the system and push instant notification popups to your screen.
 - Node-Cron (Task Scheduler): An automated clock that wakes up every morning at 8:00 AM to check for employee birthdays and upcoming evaluation dates.
 - Nodemailer (Email System): Sends 6-digit MFA security codes, password setup links, and daily birthday digests straight to employee inboxes.

3. Key Terminology & Domain Glossary

To help everyone speak the same language, here is a glossary of core terms used across EIMS:

Term	Plain-English Definition
------	--------------------------

EMPLOYEE INFORMATION	The comprehensive digital record containing an employee's personal info, HR status, government IDs, and assigned department/campaign details.
DEPARTMENT/CAMPAIGN	The specific operational team, client account, or internal unit (e.g., Customer Support Campaign, Administration, Technical Support) an employee belongs to.
REMOTE ID	The secure remote desktop identification code used by IT support staff to remotely troubleshoot an employee's workstation.
User Profile	A system login account assigned to HR managers, IT staff, or administrators that gives them access to the EIMS web application.
Role	A pre-defined collection of permissions (e.g., hr_admin, it_admin, director, super_admin) assigned to a user profile.
Capability	A specific, fine-grained permission key (e.g., employees.create, employees.secrets.view, assets.edit). Roles are made up of multiple capabilities.

Capability Overrides	Special custom permissions granted to a specific individual user on top of their base role.
Provisioning Status	The current stage of an employee's IT setup (e.g., pending_hr, pending_it, completed).
Hardware Asset	Physical equipment owned by BigOutsource (laptops, desktops, monitors, accessories) tracked in the system asset inventory.
Import Staging Area	A sandbox table where bulk employee lists uploaded from Excel are reviewed, cleaned, and checked for duplicate errors before being added to the real database.
Audit Log	An automatic, tamper-proof record detailing who made a change in the system, when it happened, what field was modified, and the IP address used.
MFA OTP	Multi-Factor Authentication One-Time Password—a 6-digit security code emailed to users during login to verify their identity.

4. Key Workflows & User Journeys

Walkthrough 1: Secure Admin Registration & User Password Setup

Step 1: Admin Creates Account —▶ Step 2: System Sends Setup Email —▶ Step 3: User Sets Password & Enters MFA

1. Account Creation: An administrator navigates to User Management and enters a new employee's name, email, department, and site. No plain-text temporary password needs to be typed.
2. Automated Link Dispatch: EIMS generates a secure, single-use security token and emails a customized invitation link (/setup-password?token=...) to the user.
3. Password Customization: The user opens the email link, types their chosen secure password, and submits. The system immediately invalidates the setup link so it cannot be reused.
4. MFA Verification: EIMS dispatches a 6-digit verification code to the user's inbox. Once typed, the user is logged in and redirected to the dashboard.

Walkthrough 2: Employee Onboarding & Cross-Departmental IT Provisioning

HR Creates Record —▶ IT Receives Alert —▶ IT Updates Credentials & REMOTE ID —▶ Record Fully Provisioned

1. HR Record Entry: HR fills out the EMPLOYEE INFORMATION form (name, contact details, date hired, employment status, and assigned DEPARTMENT/CAMPAIGN).
2. Automated IT Handoff: The moment HR saves the record, EIMS automatically sets the provisioning status to pending_it and fires an instant notification to all IT Administrators.
3. IT Provisioning: IT opens the employee profile and inputs:
 - Corporate BigOutsource email & LMS account.
 - PC workstation name & Windows license key.
 - REMOTE ID code for remote desktop support.
 - ESET Antivirus and ActivityWatch monitoring statuses.
4. Completion: IT saves the record. HR receives a green success notification confirming that the employee is fully provisioned and ready for work.

Walkthrough 3: Automated Milestone Alerts & Exporting Reports

8:00 AM Cron Check —▶ In-App & Email Alerts Sent —▶ Executive Downloads Customized Excel Report

1. **Daily Background Job:** Every morning at 8:00 AM, the server automatically checks all active employee records for:
 - Birthdays matching today's date.
 - Evaluation dates reaching the 1st month, 3rd month, 5th month, 6th month, or annual anniversary mark.
2. **Multi-Channel Notification:**
 - HR admins receive a festive birthday email digest titled "🎂 Today's Birthdays".
 - HR managers receive in-app notification badges listing employees due for performance evaluations.
3. **Exporting Custom Reports:** A Director or HR Admin navigates to **Reports (/reports)**, selects desired filters (e.g., "Active IT Assets" or "Department Roster"), and clicks **Export Excel**. The system generates a formatted spreadsheet download.

5. Roles & Access Control

EIMS uses a security framework called **Capability-Based Access Control (CBAC)**. Users are assigned one of 5 standard roles, each tailored to specific job responsibilities:

User Access Tier	Intended Audience	Core Capabilities & Permissions
Super Admin	System Administrators, IT Leadership	Full system control. Can manage user accounts, assign custom capability overrides, edit role definitions, configure global settings, and perform all actions across the portal.

HR Admin	HR Officers, Personnel Managers	Manage EMPLOYEE INFORMATION, employee onboarding, evaluation dates, HMO enrollments, department rosters, bulk employee imports, and daily birthday alerts. Restricted from viewing sensitive IT secrets/keys.
IT Admin	IT Support Engineers, System Admins	Manage IT provisioning fields, corporate emails, REMOTE ID tools, hardware asset inventory, software licenses, and IT export reports. Restricted from editing confidential HR personal data.
Viewer	Team Leads, Read-Only Auditors	Basic read-only access to view the public employee directory and department listings. Cannot edit records, view sensitive IT keys, or perform administrative tasks.

6. How to Request Changes or Support

Project Ownership & Maintenance

- Repository: Makixenn/bigoutsource-eims
- Lead Engineering Team: BigOutsource Technical Operations & Software Development Team

How to Request New Features

If your department identifies a business need for a new feature (e.g., a new report export, additional employee fields, or new notification triggers):

1. Submit a Feature Request: Contact the Lead Product Manager or submit an internal ticket specifying:
 - Feature Name & Goal: What problem does this solve?
 - Target Users: Which roles will use this feature (HR, IT, Directors)?
 - Acceptance Criteria: What exact steps or fields are required?

How to Report Bugs or System Issues

If you encounter an unexpected error, incorrect data calculation, or system issue:

1. Check the System Audit Logs (/logs) or browser network status if applicable.
2. Submit a bug report containing:
 - Summary of Issue: Short description of what went wrong.
 - Steps to Reproduce: What page were you on, and what button was clicked?
 - Expected vs. Actual Result: What should have happened vs. what actually occurred.
 - Screenshots / Error Messages: Include screen grabs of any red alert banners or error text.